Take the dataset that you used in Assignment 7 for POS tagging.

1. Implement a Maximum Entropy Markov Model (MEMM) on the training dataset and evaluate it on the dev and test sets. Use https://github.com/arne-cl/nltk-maxent-pos-tagger for implementing the tagger.
2. Implement a CRF based POS tagger on the training dataset and evaluate the model on the dev and test sets. Use https://sklearn-crfsuite.readthedocs.io/en/latest/tutorial.html for the implementation. Install sklearn-crfsuite package.
   The features to be used are:
   a. Prefixes Upto Length 4 (Prefix lengths 1, 2, 3, 4)
      i. If a prefix of particular length is not found, make that feature = NULL
   b. Suffixes Upto Length 4 (Prefix lengths 1, 2, 3, 4)
      i. If a suffix of particular length is not found, make that feature = NULL
   c. Context Features
      i. Previous Word
      ii. Current Word
      iii. Next Word
   d. Binary Length Feature
      i. If length of the word is <= 4, the length_feature = “LESS”
      ii. Otherwise “MORE”
3. Make a comparative study between MEMM, CRF, Naive Bayes, and HMM (that you developed in Assignment 7). Compare the performance of the models in each tag and plot the bar charts.